AP BIOLOGY TOPIC 5.4

Non-Mendelian Genetics: Linkage, Sex-Linkage & Extranuclear DNA

UNIT 5: HEREDITY (8–11% EXAM WEIGHT)

IST-1.J | LINKED GENES (CM) | SEX-LINKED | MATERNAL MTDNA

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§1. KEY TERMS & FORMAL DEFINITIONS LEXICON — TOPIC 5.4 (NON-MENDELIAN GENETICS: LINKAGE, SEX-LINKAGE & EXTRANUCLEAR DNA)

CED Core Vocabulary

Linked Genes — Genes located physically close together on the same chromosome do NOT assort independently and tend to be inherited together during meiosis.

Distance ↔ Recombination Rule — Closer genes → lower probability of a crossover between them ("stuck together" → low RF).

Sex Chromosomes in Humans — Females are XX (homogametic); Males are XY (heterogametic).

Key Hallmarks — Much more frequent in males; NO father-to-son transmission (fathers pass Y to sons, X to daughters); often skips generations via carrier mothers.

How Crossing Over Breaks Linkage — During Prophase I , homologous recombination exchanges segments between non-sister chromatids, producing a minority of recombinant gametes alongside the majority...

Farther apart genes — → higher probability of crossing over ("easier to separate" → higher RF, maxing out at 50% where they behave as unlinked).

X-Linked Recessive Traits — (e.g., red-green colorblindness, hemophilia A, Duchenne muscular dystrophy, Drosophila white eyes): Males (X^aY) are hemizygous: Having only one X chromosome ,...

X-Linked Dominant — Single copy causes trait; affected father (X^AY) passes trait to ALL daughters and NO sons .

§2. GENETIC LINKAGE & CHROMOSOME MAPPING (CM)

Violates 9:3:3:1 / 1:1:1:1

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Genetic Linkage & Chromosome Mapping (cM) ➔ **Linked Genes** ➔

How Crossing Over Breaks Linkage ➔ **Distance ↔ Recombination Rule**

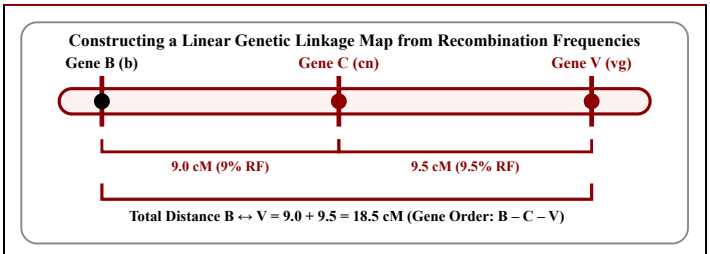
- **Linked Genes:** Genes located physically close together on the **same chromosome** do **NOT assort independently** and tend to be inherited together during meiosis.
- **How Crossing Over Breaks Linkage:** During **Prophase I**, homologous recombination exchanges segments between non-sister chromatids, producing a minority of **recombinant gametes** alongside the majority of **parental gametes**.
- **Distance ↔ Recombination Rule:**
  - **Closer genes** → lower probability of a crossover between them ("stuck together" → low RF).
  - **Farther apart genes** → higher probability of crossing over ("easier to separate" → higher RF, maxing out at **50%** where they behave as unlinked).

Recombination Frequency (RF) = (Total Recombinant Offspring / Total Offspring) × 100%

1% Recombination = 1 Map Unit (mu) = 1 Centimorgan (cM)

[AP EXAM FRQ SCORING STRATEGY] **Spotting Linked Genes in Data**

Tables: In a dihybrid test cross (**AaBb × aabb**), unlinked genes yield a **1:1:1:1** ratio (25% each). If you see **two huge classes (Parentals)** and **two tiny classes (Recombinants)** (e.g., 420, 410, 85, 85), the genes are **linked on the same chromosome**!



§4. COMPLEX ALLELIC & MULTI-GENE INHERITANCE PATTERNS

Beyond Simple Dominance

PATTERN	MECHANISM & RATIO	CLASSIC AP BIOLOGY EXAMPLES
Incomplete Dominance	Neither allele is fully dominant; heterozygote shows an **intermediate (blended)** phenotype. F_2 Phenotype = Genotype = **1 : 2 : 1**	**Snapdragon flowers:** Red (C^RC^R) × White (C^WC^W) → **Pink (** C^RC^W **)**
Codominance	Both alleles are **fully and simultaneously expressed** in the heterozygote (no blending).	**Human** I^AI^B **(Type AB) blood** expresses both A and B surface glycoproteins; roan cattle coat
Multiple Alleles	**>2 alleles** exist in the population gene pool (though each diploid organism still carries only 2).	**ABO Blood Group:** 3 alleles (I^A, I^B codominant; i recessive → 4 phenotypes: A, B, AB, O)
Pleiotropy	**1 single gene** affects **multiple seemingly unrelated phenotypic traits** across organ systems.	**Sickle cell disease** (β-globin mutation alters RBC shape, oxygen transport, blood flow, organ damage & malaria resistance); Marfan syndrome
Epistasis	Gene at one locus masks or alters phenotypic expression of a gene at a **second locus**. Modifies 9:3:3:1 to **9 : 3 : 4** (or 9:7).	**Labrador Retriever coat color:** **B/b** controls pigment (Black vs. Brown); homozygous recessive **ee** blocks pigment deposition → Yellow lab regardless of **B_**
Polygenic Inheritance	**Multiple genes** have an additive effect on a single quantitative trait → **continuous bell-curve distribution**.	Human height, skin pigmentation, eye color, intelligence

[COMMON AP EXAM FRQ TRAP] **Don't Assume Mendelian Ratios Always**

Apply: Always inspect observed offspring ratios (and run χ^2) before assuming standard 3:1 or 9:3:3:1 dominance!

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§3. SEX-LINKED INHERITANCE (X-LINKED & Y-LINKED)

Hemizygous Males

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Sex-Linked Inheritance (X-Linked & Y-Linked) ➔ **Sex Chromosomes in Humans** ➔ **X-Linked Recessive Traits** ➔ **X-Linked Dominant** ➔ **Y-Linked (Holandric)**

- **Sex Chromosomes in Humans:** Females are **XX** (homogametic); Males are **XY** (heterogametic).
- **X-Linked Recessive Traits** (e.g., *red-green colorblindness, hemophilia A, Duchenne muscular dystrophy, Drosophila white eyes*):
  - **Males (X^aY) are hemizygous:** Having only **one X chromosome**, a single recessive allele is expressed directly!
  - **Females (X^aX^a)** must inherit **two recessive alleles** (one from each parent) to express the trait; X^AX^a females are unaffected **carriers**.
  - **Key Hallmarks:** Much more frequent in males; **NO father-to-son transmission** (fathers pass Y to sons, X to daughters); often skips generations via carrier mothers.
- **X-Linked Dominant:** Single copy causes trait; affected father (X^AY) passes trait to **ALL daughters** and **NO sons**.
- **Y-Linked (Holandric):** Carried on Y chromosome (e.g., *SRY* gene) → affects **only males**; passed from **father to 100% of sons**.

[COMMON AP EXAM FRQ TRAP] **Males Cannot Be X-Linked Carriers:** Never refer to a male (XY) as a "carrier" of an X-linked trait — because males have only one X chromosome, they either show the phenotype or do not carry the allele at all!

§5. NON-NUCLEAR (CYTOPLASMIC / MATERNAL) INHERITANCE

Mitochondria & Chloroplasts

HIERARCHICAL CONCEPT & PATHWAY MAP [CAUSE ➔ MECHANISM ➔ OUTCOME]

Non-Nuclear (Cytoplasmic / Maternal) Inheritance ➔ **Genetic Basis** ➔ **Strict Maternal Transmission**

- **Genetic Basis:** **Mitochondria** and **chloroplasts (plastids)** contain their own small **circular DNA genomes** (evidence for endosymbiosis) that replicate independently of nuclear chromosomes and do **NOT** undergo Mendelian segregation.
- **Strict Maternal Transmission:**
  - During fertilization, the **female ovum (egg)** contributes virtually 100% of the zygote's cytoplasm and cytoplasmic organelles; the sperm contributes only a bare haploid nucleus.
  - **Affected Mother → 100% of her offspring** (both sons and daughters) inherit the trait!
  - **Affected Father → 0% of his offspring** inherit the trait!

[EXAMPLE & BIOLOGICAL APPLICATION]

- **Human Mitochondrial Myopathies / Leber's Hereditary Optic Neuropathy:** Mutations in mitochondrial ETC/ATP synthase genes reduce cellular ATP production (severely impacting high-energy nervous and muscle tissues) and are passed strictly from mother to all children.
- **Variegated Leaf Color in Plants (e.g., Four O'Clock *Mirabilis jalapa*):** Determined by maternal ovule chloroplast/plastid DNA, not pollen.

[CORE AXIOM / REMEMBER] **Pedigree Instant ID — X-Linked Dominant vs. Maternal Inheritance:** In **Maternal (mitochondrial)** pedigrees, an affected mother passes the trait to **every single child (100%)**, whereas an affected father passes it to **none (0%)**.

§6. SCHOLARLY SYNTHESIS: EXPERIMENTAL DESIGN, QUANTITATIVE ANALYSIS, PATHOLOGY & AP FRQ RHETORIC

High-Yield Exam Rigor

MASTER TOPIC SYNTHESIS & CAUSAL ARCHITECTURE — TOPIC 5.4 [MOLECULAR CAUSE ➔ CELLULAR MECHANISM ➔ PHYSIOLOGICAL / ECOLOGICAL EFFECT ➔ FRQ JUSTIFICATION]

Topic 5.4: Non-Mendelian Genetics: Linkage, Sex-Linkage & Extranuclear DNA ➔ **Structural / Energetic Determinants** ➔ **Regulated Pathway & Homeostasis** ➔ **Experimental Perturbation (±2 SE_x̄)** ➔ **Phenotypic / Evolutionary Outcome**

SCHOLARLY DIMENSION	COMPREHENSIVE ACADEMIC REFERENCE & MECHANISTIC SPECIFICATION
I. Experimental Design & Required Lab Controls	AP Investigation 7 Part 2 (Sordaria fimicola Meiotic Crossing Over) & Drosophila/Fast Plants Crosses: Counting 4:4 non-recombinant vs. 2:2:2:2 / 2:4:2 recombinant asci in Sordaria perithecia to determine gene-to-centromere map distance (% Crossover = [Recombinant asci / Total asci] × 100; Map distance in cM = % Crossover / 2 because only 2 of 4 chromatids in a tetrad cross over). **Null Hypothesis (H_0):** State explicitly that the independent variable treatment has *no statistically significant effect* on the dependent variable; any observed difference is due to random sampling chance alone.
II. Quantitative Formulas & Graph Interpretation	Quantitative Formulas & Statistical Testing: Chi-Square Goodness-of-Fit: $\chi^2 = \Sigma [(O - E)^2 / E]$, with Degrees of Freedom df = (number of phenotypic classes − 1). At $\alpha = 0.05$: if $\chi^2 >$ critical value, REJECT H_0 (e.g., genes are autosomal linked rather than assorting independently 9:3:3:1); if $\chi^2 \leq$ critical value, FAIL TO REJECT H_0. Recombination Frequency: RF = (Total Recombinants / Total Offspring) × 100% = Map Units (cM). **Error Bar Rule (±2 SE_x̄):** If the ±2 SE_x̄ error bars of two treatment groups *do not overlap*, there is a statistically significant difference between means ($p < 0.05$; reject H_0).
III. Pathology, Mutation & Disruption Case Study	Clinical & Chromosomal Disruption Case Study: Meiotic Nondisjunction (failure of homologous chromosomes to separate in Anaphase I, or sister chromatids to separate in Anaphase II) yields aneuploid gametes (n + 1 or n − 1). Fertilization produces Trisomy 21 (Down Syndrome, 47, +21), Klinefelter Syndrome (47, XXY), or monosomic Turner Syndrome (45, X0). Mitochondrial myopathies (e.g., Leber Hereditary Optic Neuropathy) exhibit strict maternal inheritance because the ovum contributes virtually all zygotic cytoplasm and mitochondria.
IV. AP FRQ Task Verb Scoring Templates	• **DESCRIBE / IDENTIFY:** State the direct trend, value, or structural component with exact units from the figure. • **EXPLAIN (How/Why):** Connect cause to effect via a 3-link chain: *"[Change in X] causes [molecular/cellular mechanism Y], which results in [observable outcome Z]."* • **PREDICT:** State the specific directional change (increase, decrease, or remain constant) when the pathway is perturbed. • **JUSTIFY:** Pair quantitative data/evidence from the prompt with the underlying biological principle from Topic 5.4 to prove why the claim holds.

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